MIL-S-42022(DM)
23 September 1986

MILITARY SPECIFICATION

SCISSORS, WIRE CUTTING

This specification is approved for use within the Defense Personnel Support Center and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an angular head wire cutting scissors.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specification.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-up.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

Federal Test Method Standard No. 151 - Metals; Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DPSC Drawing 24536 - Scissors, Wire Cutting, Angular

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

E 140 - Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

F 899 - Stainless Steel Billet, Bar and Wire for Surgical Instruments.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

UNITED STATES PHARMACOPEIA (USP)

The United States Pharmacopeia

(Application for copies should be addressed to Mack Publishing Company, Easton, PA 18042).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Scissors shall be manufactured from Class 4 martensitic corrosion-resistant steel. The chemical composition of the steel shall be in accordance with ASTM F899 for the class and type used. Composition shall be tested as specified in 4.5.1.

3.2 Hardness. Scissors shall be hardened to a Rockwell hardness of HRC 48-53. Hardness shall be tested as specified in 4.5.2.

3.3 Style and design. Scissors shall be in accordance with DPSC Drawing 24536. Both distal tips shall be blunt.

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3.4 Finish. The finish shall be satin or mirror and satin, as defined in Federal Specification GG-I-526. The finish shall be uniform and free from cracks, pits, crevices, burrs, and grind or tool marks.

3.5 Passivation. The scissors shall be properly passivated.

3.6 Markings. The scissors shall be permanently marked with the manufacturer's name or registered trademark and the word "stainless." In addition, instruments of foreign manufacture shall bear the name of the country of manufacture.

3.7 Performance. The scissors shall cut austenitic corrosion resistant steel wire sutures, size 3-0 USP (0.01 inch outside diameter) to size 10 USP (0.05 inch outside diameter). Performance shall be tested as specified in 4.5.3.

3.8 Resistance to corrosion. Scissors shall show no signs of corrosion or plating of copper sulfate when tested as specified in 4.5.4. Slight signs of corrosion in the serrations or internal (unexposed) joint faces shall be disregarded.

3.9 Workmanship. The scissors shall be free from any defects which affect their appearance or utility.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, or does it commit the Government to acceptance of defective material.

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4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties), of the item.

4.2.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.2.2 Certificate of quality. Certificate of quality supplied by the manufacturer of the steel may be furnished in lieu of actual performance of composition testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.3 Sampling.

4.3.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. Unit of product for examination purpose shall be one scissors.

TABLE I. Sampling criteria for examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.3.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purposes shall be one scissors. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

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TABLE II. Sampling criteria for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Composition	3.1	4.5.1	1/	2/
Hardness	3.2	4.5.2	S-2	1.0
Performance	3.7	4.5.3	S-2	1.0
Corrosion	3.8	4.5.4	S-2	1.0

1/ A sufficient amount of raw material shall be tested.

2/ The lot shall be rejected if test requirements are not met.

4.4 Examination.

4.4.1 Examination of scissors. The scissors shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any scissors in the sample containing one or more defects shall be rejected and if the number of defective scissors in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined
Major	
101	Scissors not of the specified design or dimensions.
102	Scissors not free from cracks, pits, burrs, sharp edges, scale, rough areas or corrosion.
103	Finish of scissors not uniform and not as specified.
104	Scissors has grind or tool parts.
105	Identification markings not present, not complete or not legible.
Minor	
201	Scissors not free from dirt, polishing material or foreign matter.

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4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Composition. Raw material shall be tested for composition in accordance with Federal Test Method 111 or 112 of Federal Test Method Standard Number 151 (see 4.2.2).

4.5.2 Hardness test. Hardness shall be determined in accordance with ASTM Standards E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Standard E 140.

4.5.3 Performance test. The scissors shall be tested in accordance with the scissors performance test of Federal Specification GG-I-526. Two separate tests shall be performed.

The test material for the first test shall be an austenitic corrosion-resistant steel wire of 0.010 ± 0.005 inches outside diameter (see Note 6.4). The cuts shall be made with the serrated portion of the scissor jaws.

The test material for the second test shall be an austenitic corrosion-resistant steel wire of 0.0495 (minimum) inches outside diameter. The cuts shall be made with the notched portion of the scissor jaws.

Each test shall consist of three separate cuts. The wire shall not bind or snag in the scissor jaw, nor shall the cut cause the wire tip to bend. After cutting, the jaws shall be examined. There shall be no nicks or damage to the scissors caused by cutting.

4.5.4 Corrosion tests. Scissors shall be tested for resistance to corrosion as specified in 3.8 and 4.5.4.1 and 4.5.4.2.

4.5.4.1 Boil test. The scissors shall be tested in accordance with the resistance to boil test of Federal Specification GG-I-526.

4.5.4.2 Copper sulfate test. The scissors shall be tested in accordance with the resistance to copper sulfate test of Federal Specification GG-I-526.

4.6 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The scissors shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit Package. Each pair of scissors shall be packaged in accordance with MIL-P-116, Method IC-3.

5.1.1.4 Intermediate package. Twelve unit packages shall be packaged in a box conforming to PPP-B-566, PPP-B-636, Class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for scissors shall conform to the MIL-STD-794 requirements for level C.

5.1.2.2 Intermediate package. Twelve unit packages shall be packaged in accordance with the requirements of MIL-STD-794 for level C.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the box specification.

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5.2.3 Level C. Twelve intermediate packages (144 units) shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Scissors packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Scissors packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.6.

6. NOTES

6.1 Intended use. The scissors are intended for use in cutting wire sutures, wire, and wire mesh in surgical procedures.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National Stock Number.

6.3 National Stock Number Coverage. This specification is intended to cover the following National Stock Number:

National Stock No.	Item Identification
6515-01-151-1822	SCISSORS, WIRE CUTTING, ANGULAR, 4-3/4 Inches

6.4 Performance test material. Material for the small wire performance test may be purchased as size 3.0 (0.01 o.d.) suture wire from:

Davis and Geck - Danbury, CT
Ethicon, Inc. - Somerville, NJ

6.5 Subject term (key word) listing.

Wire cutters
Wire nippers

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
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FDA-MPQAS

Project No. 6515-3884

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-42022 (DM)	SCISSORS, WIRE CUTTING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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